US Overdose Death Trends (2015–2025): Forecasting & Early Warning System

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CDC VSRR Provisional Drug Overdose Death Counts

Problem Context

Overdose deaths are reported monthly across states but identifying which regions or drug categories are about to rise is challenging. The goal is to move from tracking past trends to anticipating near-term shifts.

Primary Dataset

Dataset Name	Source	URL	Rows	Key Columns
VSRR Provisional Drug Overdose Death Counts	Centers for Disease Control and Prevention (CDC), National Center for Health Statistics	<link>	81,270	State (Geographic unit) Year & Month (Time index) Indicator (Drug category) Data Value (12-month rolling death count)

Why This Data Matters

- 10 years of consistent, state-level monthly overdose data
- Enables comparison across states and drug categories over time
- Limitation: 12-month rolling totals smooth short-term spikes

Use Cases

Use Case 1: State public health departments → Monitor overdose trends and identify rising drug categories within their state
Use Case 2: Federal health agencies → Compare states to prioritize funding and intervention efforts
Use Case 3: Policy analysts → Evaluate how overdose patterns shift over time across substances (e.g., opioids vs stimulants)
Use Case 4: Early warning system developers → Use historical trends to forecast short-term increases in overdose deaths

Initial Exploration: What We Found

Dataset Overview

- Rows: **81,270**
- Columns: **12**
- Date range: **2015 to 2025**
- Key ID column: **State + Year + Month + Indicator**

Data Quality Issues Found

- Missing values:
  - Data Value: 14,603 (~18%)
  - Predicted Value: 28,476 (~35%)
- Duplicates: None found in initial exploration.
- Outliers: Right-skewed distribution; max death count = 999
- Format issues: Data Value stored as object; no unified datetime column

Key Observations

- Death counts are uneven across states, with a few states consistently reporting higher totals.
- Data completeness is high overall, though some values are suppressed or missing.
- Trends differ across drug categories, suggesting shifting drivers over time.

Sample Data Preview (Optional: paste screenshot or describe key columns)

Each row represents a state, month, and drug category, with a 12-month rolling overdose death count.

Key columns:
- **State:** Two-letter state abbreviation (e.g., NY, CA, OH)
- **Year & Month:** Time reference for the 12-month ending period
- **Indicator:** Drug category (e.g., Synthetic opioids, Cocaine, Psychostimulants)
- **Data Value:** 12-month rolling total of overdose deaths for that state and drug category

Questions That Emerged

- Why do a small number of states account for disproportionately high overdose totals compared to others?
- Do trends across drug categories move together, or do certain substances rise independently?

The Plan: Cleaning, EDA & Enrichment

Data Cleaning Plan

Based on issues found:
• Convert Data Value to numeric
• Create proper datetime from Year + Month
• Keep missing values (no naive imputation)
• Analyze extreme values instead of removing
• Optionally merge state population for per-capita rates

EDA Questions to Investigate

• Distribution: death counts
• Relationship Deaths over time by drug category
• Comparison: State-level comparisons
• Trends 2015 - 2025 trend patterns
• Hypothesis: synthetic opioids drive recent increases

Additional Data to Enrich

• <Dataset 1>
 - Why: <what it adds>
 - Join on: <key column>
• <Dataset 2>
 - Why: <what it adds>
 - Join on: <key column>

Next Steps

- Normalize death counts per capita using population data
- Analyze trend differences across states and drug categories
- Explore forecasting approaches for 1–3 month prediction

Problem Statement:

Drug overdose deaths remain a major public-health crisis in the United States, and the pattern of deaths has changed rapidly over the last decade due to the spread of synthetic opioids (especially fentanyl), rising stimulant-related deaths, and increasingly lethal drug combinations. Because reporting delays and provisional data issues can hide emerging trends, public-health decision makers need a reliable way to monitor changes at the state level and detect early warning signals. This project analyzes CDC provisional overdose death counts from 2015-2025 with special attention to synthetic opioids, psychostimulants, and cocaine, with the long-term objective of building a forecasting and early warning system.

Use Case 1: State Public Health Departments

State public health departments can track overdose patterns to detect rising trends of overdose-related deaths, which are often related to drugs. This would help implement intervention strategies to curb the problem.

Use Case 2: Federal Health Agencies

Federal agencies can analyze overdose patterns to understand variations across different states, helping to allocate resources more efficiently.

Use Case 3: Policy Analysts and Researchers

Policy analysts can study trends related to overdose patterns, including trends related to different types of drugs, such as opioids and stimulants.

Use Case 4: Early Warning System Developers

Overdose patterns can be analyzed to build a system that predicts short-term trends of overdose-related deaths, enabling a more proactive approach to handle the problem rather than a reactive one.

This problem impacts stakeholders in multiple real-world situations:

1. State and county health departments must decide where to deploy harm-reduction resources (e.g., naloxone distribution, outreach programs) when overdose risk spikes.
2. Hospitals and EMS systems face surges in overdose-related emergencies and need early indicators for staffing and preparedness.
3. Community organizations and policymakers need timely evidence to support prevention campaigns and targeted interventions, especially when a new drug wave spreads geographically.

Quantification of significance

1. People affected (Mortality): The U.S. had recorded 79,384 deaths due to drug overdoses in the year 2024 (final mortality data). The CDC also found that "the rate of death from drug overdoses declined by 26.2% from 2023 to 2024 (from 31.3 to 23.1 per 100,000 population). This indicates the issue is large in scope and changes over time."

2. Frequency/Operational Relevance: Overdose deaths are monitored on a monthly basis as provisional counts with 12-month rolling totals; therefore, there is ongoing decision-making on a month-to-month basis.

How our approach differs and its scalability:

Unlike static annual summaries, this work is designed for **near-term (1-3 month) forecasting** using **monthly rolling totals** and category-specific signals (synthetic opioids, psychostimulants, cocaine). The approach scales nationally because the dataset is structured consistently across states and time, allowing the same pipeline to run for all jurisdictions with minimal manual intervention.

DATA SOURCES:

We use the Centers for Disease Control and Prevention (CDC) dataset titled "VSRR Provisional Drug Overdose Death Counts (Monthly, 12-Month Rolling Totals)", downloaded on February 14, 2026. This dataset provides provisional overdose death counts at both the state and national levels and includes multiple drug category indicators such as synthetic opioids, psychostimulants, cocaine, and other relevant substances. In addition to death counts, the dataset contains important metadata fields including reporting completeness percentages and pending investigation rates, which are critical for assessing data reliability.

The raw file used in this project is:

VSRR_Provisional_Drug_Overdose_Death_Counts_20260214.csv

The dataset contains more than 50,000 records, satisfying the minimum size requirement for the project. The richness of categorical indicators, temporal structure (monthly data), and reporting quality metrics makes it well-suited for cleaning, exploratory data analysis, and future time-series forecasting tasks.

Dataset Portal:

https://data.cdc.gov/

Dataset link:

https://data.cdc.gov/National-Center-for-Health-Statistics/VSRR-Provisional-Drug-Overdose-Death-Counts/xkb8-kh2a/about_data

DATA CLEANING:

1) **Load raw CDC CSV**
Why: establish a single raw source of truth.
How: pd.read_csv() on the raw dataset file.

2) **Drop rows with missing values**
Why: missing metadata or death counts can break aggregations and reduce interpretability.
How: df.dropna().

3) **Remove duplicate records**
Why: duplicates inflate death counts and distort trends.
How: df.drop_duplicates().

4) **Reset index after row drops**
Why: ensures clean indexing for downstream operations and reproducible outputs.
How: reset_index(drop=True).

5) **Create a proper time variable ("Date")**
Why: time-series EDA/forecasting requires a standard date column, not separate year/month.
How: pd.to_datetime(Year + '-' + Month + '-01').

6) **Remove unused/redundant columns**
Why: reduces noise and ensures the processed dataset contains only analysis-relevant fields.
How: drop(['Year','Month','Period','Footnote','Footnote Symbol']).

7) **Filter to fully complete records**
Why: provisional data can be incomplete; restricting to 100% complete improves reliability of trend analysis.
How: keep rows where Percent Complete == 100.

8) **Filter out high "pending investigation" records**
Why: high pending investigation suggests unstable counts that may change substantially later.
How: keep rows where Percent Pending Investigation < 0.3. 0.3 is the parameter above that we have ignored those records because of the reliability issues.

9) **Reorder columns to standardize schema**
Why: stable column order makes EDA scripts, tables, and grading checks more reproducible.
How: move Date immediately after Indicator using a column list.

10) **Rename key variable for clarity**
Why: "Data Value" is ambiguous; "Death Count" communicates meaning directly.
How: rename Data Value → Death Count.

11) **Save cleaned dataset to processed directory**
Why: separates raw vs processed and enables reuse without repeating cleaning.
How: to_csv(data/processed/cleaned_drug_overdose_deaths.csv) and create directories with os.makedirs().

Verification: We executed the full cleaning pipeline in a fresh Python environment on a teammate's machine using pip install -r requirements.txt and confirmed the processed CSV is reproduced successfully.

EDA:

1) Dataset shape and total records

2) Data types per column

3) Missing values count and missing percentage table

4) Descriptive statistics (describe())

5) Unique states count + list of states

6) Time coverage (year range, months/periods)

7) Unique drug indicators + frequency counts

8) Non-null "Data Value" coverage and numeric summary (min/max/mean/median)

9) Visual: completeness by year

10) Visual: top states by record count

11) Visual: indicator distribution

12) Visual: histogram of death counts (log scale)

13) Visual: average death count by year

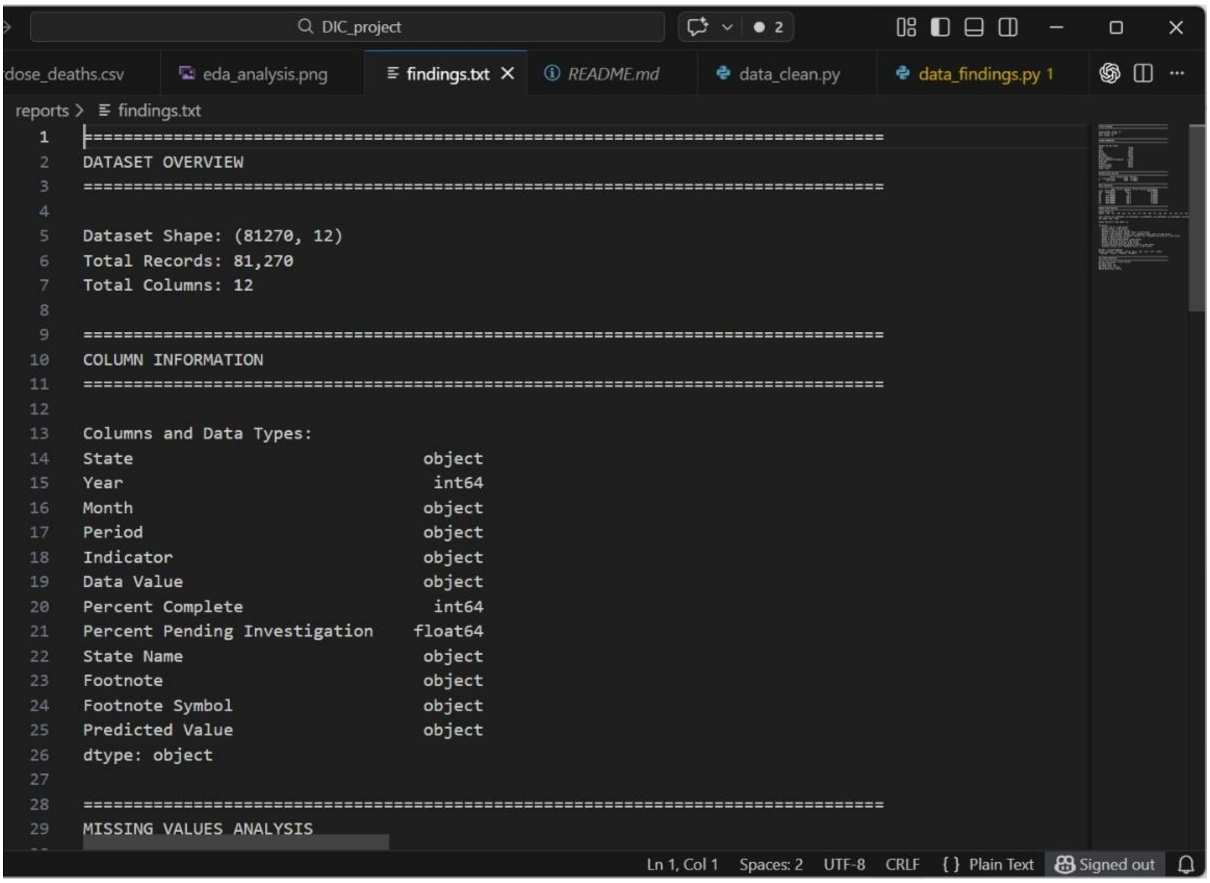
14) Visual: average percent complete by year

15) Visual: top states by total death count

16) Visual: boxplot of death counts by top indicators

17) Visual: data availability by month

EDA showed that data completeness and pending investigations vary across records, which motivated filtering to Percent Complete == 100 and limiting Percent Pending Investigation to improve reliability of trend interpretation before modeling. We also renamed Data Value to Death Count for interpretability.

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================================================================================
DATASET OVERVIEW
================================================================================

Dataset Shape: (81270, 12)
Total Records: 81,270
Total Columns: 12

================================================================================
COLUMN INFORMATION
================================================================================

Columns and Data Types:
State                             object
Year                               int64
Month                             object
Period                            object
Indicator                         object
Data Value                        object
Percent Complete                   int64
Percent Pending Investigation    float64
State Name                        object
Footnote                          object
Footnote Symbol                   object
Predicted Value                   object
dtype: object

================================================================================
MISSING VALUES ANALYSIS
```

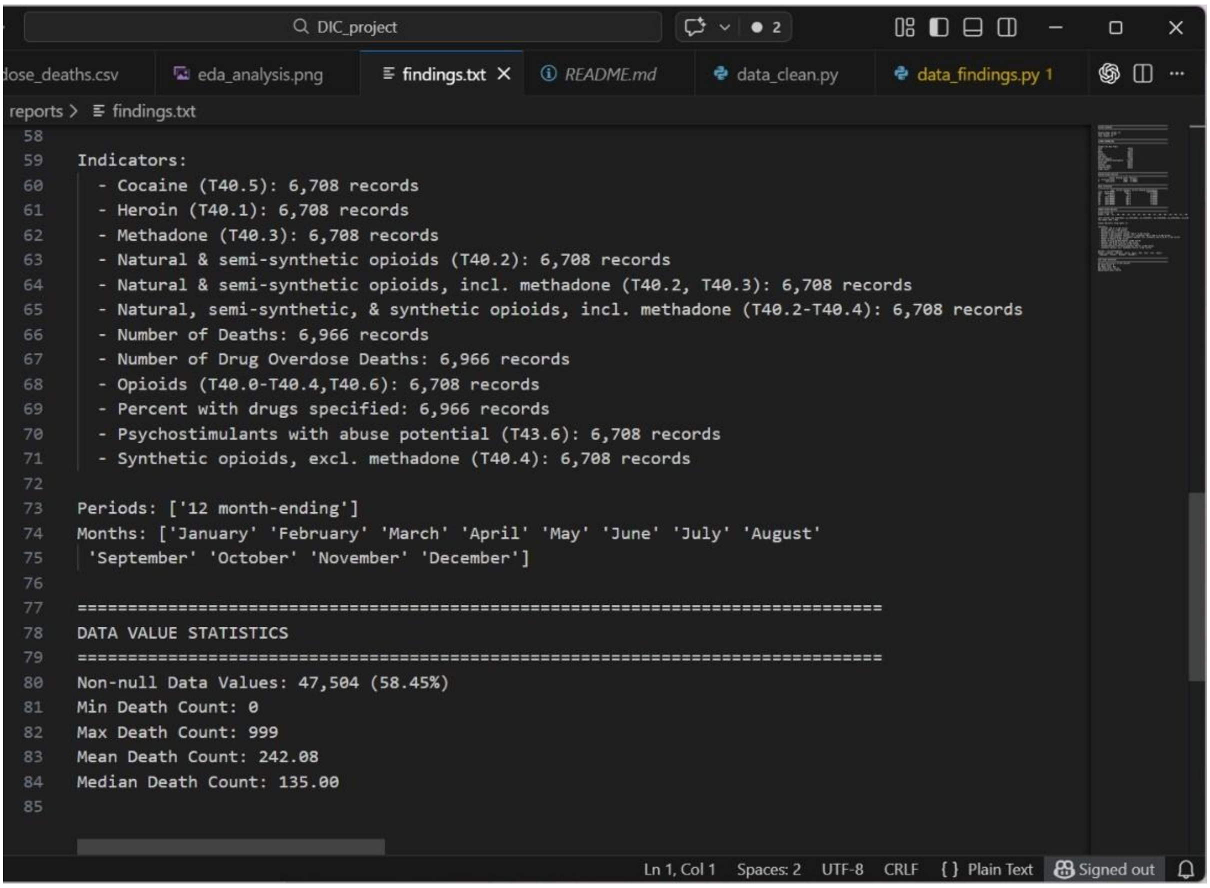
```
MISSING VALUES ANALYSIS
================================================================================
               Column  Missing Count  Missing %
11  Predicted Value          28476   35.03876
5        Data Value          14603   17.96850

================================================================================
BASIC STATISTICS
================================================================================
                Year  Percent Complete  Percent Pending Investigation
count  81270.000000           81270.0                   81270.000000
mean    2019.883721             100.0                       0.141606
std        3.104474               0.0                       0.276498
min     2015.000000             100.0                       0.000000
25%     2017.000000             100.0                       0.016918
50%     2020.000000             100.0                       0.057830
75%     2023.000000             100.0                       0.163102
max     2025.000000             100.0                       2.783646

================================================================================
UNIQUE VALUES ANALYSIS
================================================================================
Unique States: 54
States: ['AK', 'AL', 'AR', 'AZ', 'CA', 'CO', 'CT', 'DC', 'DE', 'FL', 'GA', 'HI', 'IA', 'ID', 'IL', 'I

Years Covered: [np.int64(2015), np.int64(2016), np.int64(2017), np.int64(2018), np.int64(2019), np.ir
Year Range: 2015 - 2025

Unique Indicators (Drug Types): 12
```



```
Indicators:
  - Cocaine (T40.5): 6,708 records
  - Heroin (T40.1): 6,708 records
  - Methadone (T40.3): 6,708 records
  - Natural & semi-synthetic opioids (T40.2): 6,708 records
  - Natural & semi-synthetic opioids, incl. methadone (T40.2, T40.3): 6,708 records
  - Natural, semi-synthetic, & synthetic opioids, incl. methadone (T40.2-T40.4): 6,708 records
  - Number of Deaths: 6,966 records
  - Number of Drug Overdose Deaths: 6,966 records
  - Opioids (T40.0-T40.4,T40.6): 6,708 records
  - Percent with drugs specified: 6,966 records
  - Psychostimulants with abuse potential (T43.6): 6,708 records
  - Synthetic opioids, excl. methadone (T40.4): 6,708 records

Periods: ['12 month-ending']
Months: ['January' 'February' 'March' 'April' 'May' 'June' 'July' 'August'
 'September' 'October' 'November' 'December']

================================================================================
DATA VALUE STATISTICS
================================================================================
Non-null Data Values: 47,504 (58.45%)
Min Death Count: 0
Max Death Count: 999
Mean Death Count: 242.08
Median Death Count: 135.00
```

Dead Ends:

1)One of the challenges that arose during this process was attempting to accurately calculate true month-wise overdose fatalities based on the values reported within the dataset. The CDC dataset reports fatalities as a "12-month rolling total," meaning each data point is a cumulative total of fatalities that occurred within a trailing 12-month period rather than a specific month's fatalities. The initial attempt to "decompose" this data to calculate month-wise fatalities was abandoned, as this process proved to be unstable and resulted in an inaccurate interpretation of results, as each month's fatalities are not able to be recovered uniquely within a rolling total dataset. As a result, this data was left as a "rolling total" for further trending.

2) State-Level Drug Type Correlation Matrix

An attempt was made to calculate a correlation matrix for each drug type across state data to understand which drugs clustered together and could offer a better understanding of overdose patterns. However, all drug types had uniformly high correlation values across state data, which proved to be misleading. This was because, for each state, more populous states had a higher total death rate for all drugs, making correlation values less reliable for this process.

3) Footnote Symbol as a Data Quality Filter

The dataset includes footnote symbols indicating suppressed or underreported records. We hypothesized that these symbols could help identify unreliable observations. Although records marked with “**” showed slightly higher missing values than those marked with “*”, the difference was minimal, and every row contained one of the two symbols. As a result, the footnote variable lacked discriminative power and could not be used as an effective data quality filter.

4) Percent Pending Investigation as a Missing Data Predictor

We examined whether higher percentages of pending investigations were associated with missing death counts. While rows with missing values showed slightly higher pending investigation percentages, the difference was too small to reliably distinguish incomplete records. Missingness was primarily driven by CDC suppression rules and reporting delays rather than pending investigation levels, making this variable ineffective for filtering purposes.

Surprise Findings:

1)A surprise finding was how uneven data availability and completeness can be across time (some years/months have much higher non-null coverage than others). This changed our approach by making data reliability a first-class consideration: we filtered for fully complete records and limited pending investigations before creating trend-based insights, so that patterns we observe are less likely to be artifacts of reporting delay.

Surprise Finding 2: Drug Correlations Reflected Population Size

We expected distinct drug types to show different state-level patterns. Instead, most substances showed strong correlations (0.72–0.85), indicating that larger states consistently reported higher death counts across all drug categories. The correlations reflected population size rather than drug-specific dynamics. This revealed that raw counts obscure structural differences and that per-capita normalization would be necessary to isolate meaningful substance-specific trends.

Surprise Finding 3: Indicator Categories Are Structurally Overlapping

When attempting to rank drug indicators by average death count, we discovered that many categories are nested within broader ICD classifications (e.g., heroin and methadone are included within "Opioids"). Because these categories are not mutually exclusive, direct comparisons lead to double counting. This finding forced us to treat indicators as overlapping classifications rather than independent groups.

Design decisions:

Building on the cleaned and reliability-filtered dataset developed in Phase 1, the long-term objective of this project is to construct a state-level overdose forecasting and early warning framework. The system is designed to operate using monthly rolling totals rather than reconstructed monthly counts, consistent with the dataset's structure and reporting limitations.

The proposed framework will:

- **Generate short-term (1–3 month ahead) forecasts of overdose deaths at the state level**
- **Model category-specific trends for:**
    - **Synthetic opioids (including fentanyl)**
    - **Psychostimulants**
    - **Cocaine**
    - **Total drug overdose deaths**
- **Produce an interpretable state-level risk ranking based on predicted short-term increases**

- **Provide explainability through feature importance analysis to clarify the drivers of elevated risk**

This approach reflects key design decisions made in Phase 1, including the use of rolling totals, reliability filtering (percent complete and pending investigation thresholds), and careful treatment of overlapping indicator categories. The forecasting framework is therefore grounded in structurally consistent, reproducible, and policy-relevant data.

Reference:

[1] Seth P, Baldwin GT, Davis NL, Jones CM. Clarifying CDC's Efforts to Quantify Overdose Deaths. Public Health Rep. 2023 Sep-Oct;138(5):721-726. doi: 10.1177/00333549221123586. Epub 2022 Oct 1. PMID: 36184930; PMCID: PMC10467501.